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Combining deep neural network and bibliometric indicator for emerging research topic prediction

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ABSTRACT

Predicting emerging research topics is important to researchers and policymakers. In this study, we propose a two-step solution to the problem of emerging topic prediction. The first step forecasts the future popularity score, a novel indicator reflecting the impact and growth, of candidate topics in a time-series manner. The second step selects novel topics from the candidates predicted to be popular in the first step. Terms with domain characteristics are used as candidate topics. Deep neural networks, specifically LSTM and NNAR, are applied with nine features of topics to predict popularity score. We evaluated the models and five baselines on two datasets from two perspectives, i.e., the ability to (1) predict the correct indicator value and (2) reconstruct the optimal ranking order. Two types of training strategies were compared, including a global strategy that trains a model with all topics and two local strategies that train separate models with different groups of topics. Our results show that LSTM and NNAR outperform other models in predicting the value of popularity score measured by MAE and RMSE, while LightGBM is a competitive baseline in ranking the topics in terms of NDCG@20. The performance difference of global and local strategies is not significant. Emerging topics predicted by our approach are compared with those by other methods. A qualitative assessment on nominated emerging topics suggests topics nominated by machine learning methods are more alike than those by the rule-based model. Some important topics are nominated according to a preliminary literature analysis. This study exploited the strengths of both machine learning and bibliometric indicator approaches for emerging topic prediction. Deep neural networks are applied where objective optimization target can be defined and measured. Bibliometric indicator offers an efficient way to select novel topics from candidates. The hybrid approach shows promise in considering various characteristics of emerging topics when making predictions.

1. Introduction

Identifying emerging research topics is of great interest to both funding agencies and individual researchers, as it can provide valuable insights into promising research directions and help to optimize the allocation of research funds (Small, Boyack, & Klavans, 2014; Wang, 2018). Since the pioneering studies on mapping science and detecting emergence (Price, 1965; Small & Griffith, 1974),

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this field has attracted extensive attention from policymakers and scholars of various disciplines. For instance, the Foresight and Understanding from Scientific Exposition (FUSE) program funded by the Intelligence Advanced Research Projects Activity (IARPA) in 2011, was one of the state-funded programs aiming to identify emerging topics from scientific, technical, and patent literature (McKeown et al., 2016; Meyers et al., 2018; Small et al., 2014).

Considerable effort has been made to establish frameworks and methodologies for the definition (Goldstein, 1999), characterization (Cozzens et al., 2010; Guo, Weingart, & Börner, 2011; Rotolo, Hicks, & Martin, 2015), and identification (Small et al., 2014; Wang, 2018) of emerging topics. However, most studies are retrospective, which focus on the characteristics of emerged topics and apply predefined rules to identify emerged topics based on historical data. These studies have improved our understanding of emerging topics and provided valuable features for subsequent predictive work, but they have not met the needs of decision-makers and scientists in forecasting. As the volume of publications grows at an increasingly rapid speed, it is important to identify and predict emerging topics. However, this should be performed in a timely manner to help scientists and governments foresee future research directions.

Predicting emerging research topics goes one step further than identifying them, but few studies have focused on this task. It seeks to forecast topics that will emerge in the future, which is more challenging. By establishing features that quantify novelty, impact, growth, and other characteristics of emerging topics, the task of emerging topic prediction has been attempted as a statistical learning problem (Asooja, Bordea, Vulcu, & Buitelaar, 2016; McKeown et al., 2016; Xu, Hao, An, Yang, & Wang, 2019). Given historical features as inputs and future indicators as the targets, a predictive model is trained and used to forecast emerging topics.

1.1. Research objectives and contributions

The main objective of this study is to predict emerging research topics in scientific fields. Novelty, growth, and impact are commonly discussed characteristics of an emerging topic (Rotolo et al., 2015; Small et al., 2014; Wang, 2018). The profile of an emerging topic is to start as a novel topic, and rapidly grow and gain impact in near future[1]. In this sense, novelty is not well-suited as a target for machine learning models to predict since it applies at the beginning not at the end. For this reason, we propose a two-step solution to emerging topic prediction. The first step constructs feature sequences of candidate topics and models them with machine learning to predict future popularity scores of candidate topics. The indicator of popularity score proposed in this study considers both growth and impact. The second step selects novel topics from ranked candidates produced from the first step. This approach ensures the novelty, growth, and impact of predicted topics. This study provides a framework and practical guidelines for future work by clarifying the problem formulation and problem-solving procedures.

The main contributions of this study are as follows:

- A novel indicator that reflects the impact and growth of a topic is proposed, serving as the prediction target.
- A formal definition of emerging topic prediction as a time series forecasting problem is provided, as well as a generalized framework to solve this problem, which allows predicting popularity scores of a topic at future time points.
- A hybrid approach of machine learning and bibliometric indicator to emerging topic prediction is presented.

2. Related work

The research on emerging research topics was initially conducted in the form of mapping science structure, driven by the need of finding and analyzing research fronts (Xu, Hao, An, Pang, & Li, 2020). Early studies have summarized some basic properties of emerging research topics and laid the foundation for further research. Current studies can be divided into two categories according to their research purposes, i.e., identification and prediction of emerging topics. Our work falls into the second category.

2.1. Identification of emerging research topics

Emerging research topic is a complex concept. No consistent definition of emerging research topic has been well accepted to date. Some scholars have attempted to characterize emerging research topics from theoretical perspectives. For instance, Rotolo et al. (2015) refined five attributes of emerging topics based on an extensive review of previous work, including radical novelty, relatively fast growth, coherence, prominent impact, as well as uncertainty and ambiguity. These characteristics have a solid theoretical basis but seem to be qualitatively concluded. There is still a need for practical measurements that can quantify the characteristics of emerging topics. By analyzing predefined emerging topics, some characteristics have been manifested in a data-driven manner. Guo et al. (2011) adopted burst detection algorithm (Kleinberg, 2003) and found emerging topics were preceded by word bursts, as well as the rapid growth of new authors and interdisciplinarity. Boyack, Klavans, Small, & Ungar (2014) found documents related to emerging topics concentrate in newly formed threads by constructing annual document clusters, which were linked year to year using overlapping references. The practical measurements are often related to the aforementioned attributes, but some attributes are not easy to measure. For example, uncertainty is such an attribute that is difficult to be assessed (Rotolo et al., 2015). In addition, different attributes may have associations. For instance, a topic with radical novelty could grow fast in the early years.

[1] It should be noted that the prediction of an unseen research topic is out of the scope of this study.

The characteristics derived from specific emerging topics are generalized and applied to identify emerging topics. For example, Tu & Seng (2012) used two indices based on the publication time and the number of documents related to a topic to measure its annual novelty and impact. An indicator calculated based on the combination of novelty and impact was assigned to each topic. Topics with an above-average indicator value were chosen as emerging topics.

Several works employed bibliometric approaches, which have been widely used to identify emerging research topics (Cho & Shih, 2011; Glänzel & Thijs, 2012). Small et al. (2014) constructed direct and co-citation networks based on Scopus articles, which were then clustered to represent research topics. Next, an algorithm, which rewarded clusters for having recent articles and penalized the opposite, was proposed to calculate the indicator that reflects the emergence of each topic. Topics with the highest values in the indicator were considered as emerging ones. The proposed method was validated by finding award-winning articles (e.g., Nobel Price) in the topic clusters. Similarly, Wang (2018) adopted the direct citation model but proposed different approaches to measure the growth, novelty, impact, and coherence of research topics. Instead of one compound indicator, four rules were designed to filter out emerging topics. She validated the identified topics both quantitatively and qualitatively, by analyzing their impact change and searching for related studies that also reported the same topics as emerging. Although these studies have identified emerging topics from the dataset, they rely heavily on dedicated rules. The only way to validate the results is the judgment made by human experts. Therefore, the generality and comparability of these studies are limited.

More recently, indicators and analytical frameworks serving the purposes of early identification are proposed. In Mariani, Medo, & Lafond (2019), an age-normalized PageRank and a rescaled citation count were proposed and proved more effective in ordering and recalling the most important patents. These patents were identified more quickly after issued with the proposed metrics. Complex frameworks that search for specific growth patterns were further established to identify scientific and technological breakthroughs in their early stage of development (Porter, Garner, Carley, & Newman, 2019; Winnink, Tijssen, & van Raan, 2019). The identification process is also raised to a higher level of automation thanks to the introduction of unsupervised machine learning (Altuntas, Erdogan, & Dereli, 2020; Muhlroth & Grottke, 2020). However, instead of modeling the development trajectory of a topic, these studies still defined fixed patterns or rules that were applied to filter suitable topics. While some studies have introduced machine learning models to this problem, they are mainly used to generate sophisticated features and are not involved in the decision stage. Hence, despite the use of new technologies, the literature still falls into the category of identifying emerging topics from historical data. Their objective is to detect emerging topics as early as possible. However, the experiments were still performed in a retrospective manner, and their usefulness in providing guidelines for future research and management purposes is limited.

2.2. *Predicting emerging research topics*

In the literature, emerging research topics have been detected based on hand-craft rules. However, few studies have applied statistical and machine learning models to this problem. Prior to the studies on emerging research topic prediction, the evolution of topics has been investigated by numerous researchers. This is helpful to understand the life cycle of a research topic, including how it is born and becomes emerging. Chen, Tsutsui, Ding, & Ma (2017) analyzed the evolving dynamics of research topics in the field of information retrieval through topic modeling and similarity measuring. They found that a research topic usually followed a pattern from the adjusting phase to the mature phase, with knowledge transferred between topics. Considering the influence that cited papers have on citing papers, Kim, Baek, & Song (2018) weighted the citation network with the Citation Influence Model (Dietz, Bickel, & Scheffer, 2007) and extracted the main evolution path of a biomedical topic, which can be represented and visualized by the topic distribution of papers. Hu et al. (2019) found the leading and suppressing patterns exist in the interaction between research topics, modeled by spatial autocorrelation. They proved that some topics brought additional attention to their related ones, while others suppressed nearby topics. Built upon these studies, some predictive works have been conducted. For instance, Chen, Wang, Li, & Sun (2018) proposed a correlated influenced model to predict the future topics of 11 computer science conferences, represented as bag-of-words vectors. Krenn & Zeilinger (2020) built a semantic network based on concept co-occurrences and trained a neural network model to predict future connections between concepts, which were considered as possible research trends. These studies have provided a variety of useful features and models for the solution to emerging topic prediction.

A few recent studies have attempted to forecast the future status of topics for predicting emerging topics. Asooja et al. (2016) applied linear and polynomial regression to predict the future trend of research topics in terms of their growth, measured by scaled TF-IDF values. Through the comparison on Spearman's correlation and root mean squared error (RMSE), they also found that individually modeling the patterns of each topic produced better results than the global model. McKeown et al. (2016) made use of the full-text features and Logistic regression model to forecast the changes in the impact of research topics, measured by the exponentially weighted average of related document counts. The model performance was evaluated through Pearson's and Spearman's correlation coefficients. More recently, Xu et al. (2019) generated candidate topics and quantified their emerging properties (growth, influence, and novelty) using the Dynamic Influence Model and Citation Influence Model. A support vector regressor was employed to generate predictions for each property. Emerging topics were then selected from the candidates according to whether their predicted values of emerging properties were higher than average. These topics were validated through literature study.

Different from the identification research, models adopted by predictive studies are responsible for making decisions on which topics are emerging, or predicting future values of the emerging indicator. Their solution is to model the relationship between input features and the target indicators. However, this approach to emerging topic prediction has not yet been mathematically formulated. Since the ultimate goal is to predict emerging topics with essential attributes, the most challenging issue is how to embed the attributes into predictive models. The target indicator should reflect the attributes of emerging topics. Due to the measurability and complexity of the attributes, the above methods only consider parts of the attributes as a practical and feasible solution (Asooja et al., 2016;

McKeown et al., 2016; Xu et al., 2019). Alternatively, the input features of predictive model can take the attributes into consideration. In addition, a post-selection criterion with more metrics, or even expert wisdom, can be applied to further filter the candidates of emerging topics, as in those studies of the identification task. Therefore, our work defines predicting target indicator as a time series forecasting problem, which enables a finer-grained prediction on not only what research topics will become emerging but also when (Khatibi, Belém, Couto da Silva, Almeida, & Gonçalves, 2020). Deep neural networks are applied to handle the non-linear development pattern of research topics.

3. Methodology

To predict emerging research topics with the characteristics of novelty, growth, and impact, we propose a two-step solution involving machine learning approaches and bibliometric instruments. The overall framework is shown in Fig. 1. After generating candidate research topics, the first step is formalized as a time series forecasting problem, which predicts the popularity score, an indicator based on growth and impact, of a topic based on its historical observations. The second step uses a bibliometric indicator to select novel topics from the results of the first step. Our method exploited the strengths of both machine learning and bibliometric indicator approaches.

3.1. Candidate topic generation

Although clustering (Wang, 2018) and topic modeling approaches (Xu et al., 2019) are commonly used to generate candidate topics, the generated clusters/topics typically need to be labeled for interpretation. These labels are given in the form of terms by analyzing the documents in the clusters, which often relies on expert knowledge (Jeong, Park, & Yoon, 2019; Small et al., 2014). This study adopts term extraction to generate candidate topics since it requires less human intervention, allowing for a larger number of topics to be considered.

Specifically, we use Termolator, a high-performing term extraction program developed by Meyers et al. (2018), to generate the research topics. Given the documents in the research area of interest as a foreground dataset and those about a wide range of research areas as background corpus, representative noun phrases are extracted, normalized, and scored by Termolator. It gives higher scores to the noun phrases that are more related to the foreground dataset (i.e., domain-specific) than the background corpus, measured by a set of statistical indicators. This program is capable of extracting terms with domain characteristics, e.g., candidates containing out-of-vocabulary words and technical adjectives, with an accuracy of about 85%. Top-rated terms are kept as candidate topics to be predicted.

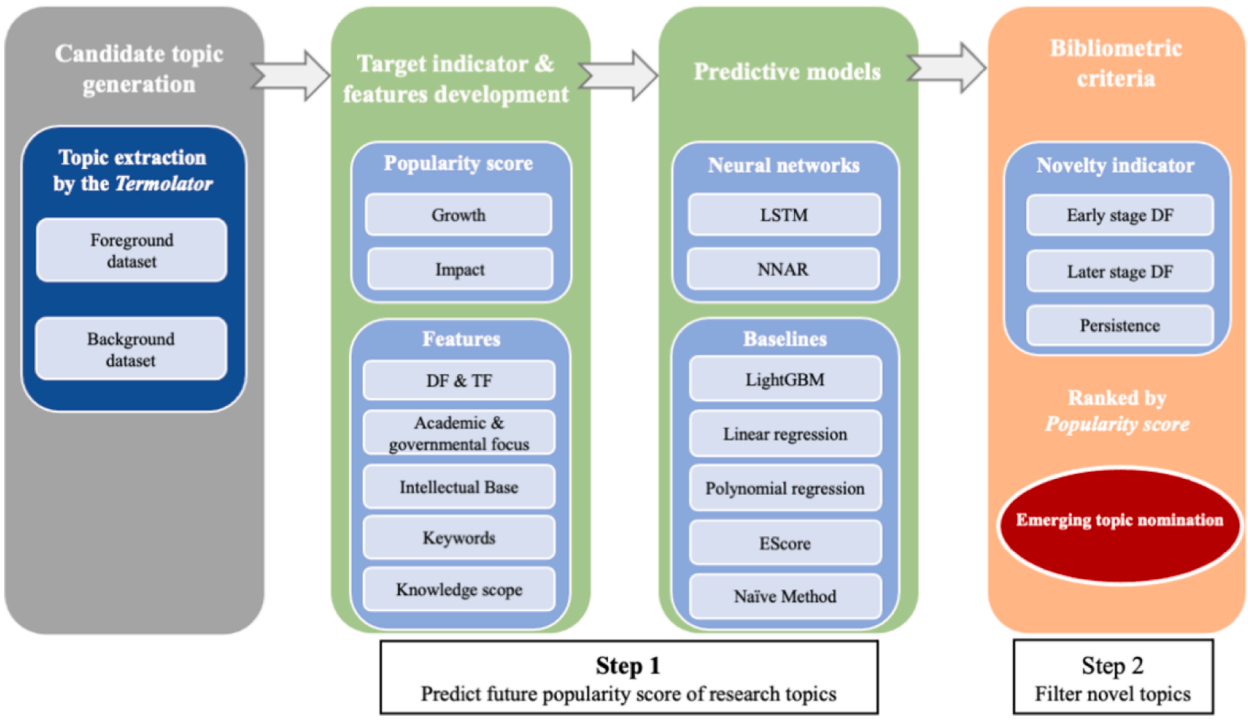


Fig. 1. Methodological framework for emerging topic prediction.

3.2. Predicting popularity scores of research topics

3.2.1. Problem formulation of predictive model

The first step of emerging research topic prediction is to forecast future popularity scores of research topics based on historical observations. This can be formulated as a time series forecasting problem as follows:

We define $X = \{X_1, X_2, ..., X_n\}$ as the set of feature time series of n research topics. A feature time series $X_i \in \mathbb{R}^{t \times m}$ is a $t \times m$ matrix with m features in t time steps (e.g., years). We also define $Y = \{Y_1, Y_2, ..., Y_n\}$ as the set of target time series of n research topics. A target time series $Y_i \in \mathbb{R}^{t' \times m'}$ is a $t' \times m'$ matrix with m' target indicators in t' time steps. Given the feature time series X as the model input, we aim to learn a mapping function f such that $\widehat{Y} = f(X)$ resulting in an accurate prediction of Y. In other words, the model is trained to predict target time series in the next t' time steps based on the feature time series from the past t time steps.

We model emerging topic prediction as a multivariate ($m > 1$) multi-step ($t' > 1$) time series forecasting problem in this study, predicting single output (m'=1, i.e., popularity score) for each topic in each year. In this case, the target time series Y_i degenerates to a vector with t' dimensions. More generally, we regard it as a regression problem since the popularity score is a numerical target. We also discuss the classification form of this problem with categorical prediction targets in the Discussion section.

3.2.2. Target indicator and input features

To measure the growth and impact of research topics, we propose one concise indicator called popularity score in which impact and growth are explicitly gauged by frequencies. They are two widely used properties to detect emergence (McKeown et al., 2016; Porter et al., 2019), which are quantified based on the number of documents related to the given topic in this study. The popularity score is calculated as follows:

$$ADF_i = DF_i + \alpha \times ADF_{i-1} \tag{1}$$

$$PopularityScore_i = \ln(ADF_i + \delta) \times \frac{DF_i + \delta}{DF_{i-1} + \delta} \tag{2}$$

The adjusted document frequency (*ADF*) of each topic (term) is calculated annually using formula (1), where DF_i represents the document frequency of given term at time *i*, ADF_{i-1} is the adjusted document frequency of this term at time *i-1*, and α is a decay factor. The *ADF* is set to the document frequency in the first year. In other words, *ADF* denotes the cumulative impact of a topic with an annual decaying mechanism (Wang, Song, & Barabási, 2013), based on the number of documents that contain specific terms. The range of α is from zero to one, which controls the extent to which historical impact should be considered in the current time. The *ADF* degenerates to the document frequency if the decay factor is set to zero; or it is equivalent to the accumulated document frequency if α is set to one. It is set to 0.9 in this study, which is a moderate decay rate that makes the topic impact half after 7 years. The effect of this parameter will be further analyzed and discussed in Section 5.1. Our model is robust across different values of decay factors.

Then, the annual growth rate of a topic is computed by dividing DF_i by DF_{i-1}. Finally, popularity score is calculated by combining the impact and growth of terms. We logarithmize the *ADF* to make impact and growth terms more balanced because *ADF* is much larger than the growth rate. A smoothing factor δ is introduced to ensure the validity of the function, even when *ADF* or *DF* equals zero. This factor is heuristically set to 1.0 in this study. A research topic with a higher popularity score indicates that it has a greater impact and grows relatively fast.

In addition to the popularity score, we leverage other input features to enhance the prediction performance. Compared with the univariate time series, input features in multivariate time series may contain additional information about the target indicator. For instance, the values of some features experience changes before the target indicator does. Therefore, forecasting on a multivariate time series tends to capture changes of the target indicator more effectively and produce more accurate results (Aboagye-Sarfo et al., 2015). Table 1 describes the input features employed in this study, which are extracted from the metadata of documents.

Among the eight features in Table 1, document frequency is the one that is most directly related to the popularity score. It has been used as the target indicator or an indicative characteristic of emerging topics by many studies (McKeown et al., 2016; Porter et al., 2019). While document frequency shows the impact of research topics at the corpus level, term frequency is a fine-grained complement

Table 1
Summary of input features employed in this research.

Feature name	Operational definition
Document frequency (DF)	Number of documents that contain the specific term
Term frequency (TF)	Number of occurrences of the specific term
Academic focus	Number of unique authors participate in this topic, disambiguated by full name and institution
Governmental focus	Number of grants supporting this topic, disambiguated by funding agency and number
Intellectual base	Number of unique references used by documents in this topic, disambiguated by first author name, venue, publication year, page number, and digital object identifier (DOI)
Keywords	Number of unique author keywords used in the related documents
KeywordPlus	Number of unique KeywordPlus used in the related document
Knowledge scope	Number of unique Web of Science categories

to it at the document level. We include them in input features as they provide direct information relevant to our target indicator.

It is shown that larger team size can lead to an increase in impact (Larivière, Gingras, Sugimoto, & Tsou, 2015), so the number of participants is considered to be an important factor in the development of research topics. Authors and their institutions are extracted from the metadata of related documents, then the number of unique authors is counted for each research topic and treated as the feature of academic focus. Similarly, the focus from governments and public agencies can boost the research on certain topics as well as their impact, which is often proxied by the number of grants associated with the papers (Gök, Rigby, & Shapira, 2016; Yan, Wu, & Song, 2018). We extract funding numbers and corresponding agencies and count the number of unique grants supporting a research topic.

In addition, the coverage of research also affects its audience size and impact (Wang, Thijs, & Glänzel, 2015). This includes the amount of knowledge absorbed and produced by the research, which can be measured based on references (Wang et al., 2015), keywords (Xu et al., 2018), and categories (Lee, Kwon, Kim, & Kwon, 2018). We also extract these types of information from the metadata.

We manage to keep the feature set concise and easy to measure. Features based on citation count and network are not included in this research due to their instability over time. The target indicator and input features of topics are measured and standardized annually, then delivered to the predictive models described below.

3.2.3. Predictive models

Stationary time series is the one that has invariant or stable statistical properties (e.g., mean and variance) over time. As the features of a research topic may experience radical changes during its development, constituting a multivariate and nonstationary time series. It is not suitable to apply classical models like exponential smoothing and autoregressive integrated moving average (ARIMA), which are originally designed for univariate and stationary data. Moreover, the non-linear relationships between the target indicator and input features are difficult to capture by these linear models. Therefore, we adopt two neural network models, including neural network autoregression (NNAR) and long short-term memory networks (LSTM), as the predictive models.

Neural network autoregression (NNAR; Taskaya-Temizel & Casey, 2005) is a feedforward neural network for time series prediction, which is similar to an autoregressive (AR) model that predicts the target at the current time based on the targets and features in the historical period. By using non-polynomial activation functions and more hidden layers, NNAR is more capable to capture nonlinearity and fit complex relations than the vanilla autoregressive models (Leshno, Lin, Pinkus, & Schocken, 1993). In addition, the neural network does not require the time series to be stationary. We use the rectified linear unit as the activation function and grid search the optimal hyperparameters (hidden layers and the number of neurons) with 5-fold cross-validation.

Despite the capability to model complex functions, NNAR does not handle temporal dependencies as features from all historical time steps are flattened and fed to it simultaneously. The weights of neurons are not shared in different layers, and the original order of sequential data is not kept either. In our cases, temporal dependencies and the order may be helpful to obtain more accurate results. As a type of recurrent neural network, LSTM explicitly handles temporal dependencies and keeps the ordinal information by its recursive structure (Hochreiter & Schmidhuber, 1997). We employ LSTM with the sequence-to-sequence structure (Sutskever, Vinyals, & Le, 2014) to better suit the problem in this research, which has been proven successful in time series forecasting tasks (Abrishami & Aliakbary, 2019). For better performance, we tune the hyperparameters with 5-fold cross-validation.

3.3. Selecting emerging topics with novelty indicator

The second step of emerging research topic prediction is to apply the bibliometric indicator to filter novel topics from the results of the first step. Among the three recognized characteristics of an emerging topic, novelty is the most difficult to quantify (Xu et al., 2019). Previous studies have attempted to represent topic novelty from the perspective of document frequency, e.g., the number of documents related to a novel topic should be relatively small in its early stage of emergence. On the other hand, novel topics should also be persistent in their later stage, otherwise they are just “one-hit wonders” (Porter et al., 2019; Wang, 2018). Adapted from the practices of Porter et al. (2019), we define three thresholds that a novel research topic should meet[2]:

(1) Early stage frequency: Its document frequency in the first three years should be lower than the 15th percentile of that of all research topics.
(2) Later stage frequency: Its document frequency in the subsequent eight years should be at least twice as that in the first three years.
(3) Persistence: It should appear in at least four years of the subsequent eight years.

Threshold (1) and (2) focus on the novelty of research topics while threshold (3) ensures that these topics are not ephemeral. Since the growth and impact of research topics have been explicitly modeled through popularity scores and the novelty through the bibliometric indicator, the emerging research topics can be nominated from the results of the second step, sorted by popularity scores.

[2] The early stage and later stage are partitioned based on a 11-year duration.

4. Experimental setups and results

4.1. Datasets

The proposed method was applied to two bibliographic datasets collected from Web of Science. The first dataset consists of 19,164 publications about Gene editing, collected via the same queries used in Huang et al. (2019). We also collected 206,573 publications from 25 journals in the subject category of Transplant, according to the 2018 Journal Citation Report. The publication year was set to the range from 2003 to 2018 for both datasets. Table 2 shows some descriptive statistics of the two datasets. The two fields are chosen since they are two important fields with many developing topics of biomedical research, which attract attention from researchers of multiple disciplines (Huang et al., 2019; Xu et al., 2019). In addition, the two datasets were obtained by different gathering methods of domain documents, i.e., query-based and journal-based.

To extract terms as candidate research topics, we used Gene editing and Transplant publications as the foreground dataset of Termolator respectively. The title, abstract, and keyword of 28,089 articles and reviews published between 2003 and 2018 in two prominent multidisciplinary journals (*Science* and *Nature*) were collected as the background dataset. Finally, 2,643 and 5,141 terms (topics) occurring in at least three documents were extracted from Gene editing and Transplant datasets. The features of these terms were calculated. The annual changes in popularity score are shown in Figs. 2 and 3 for the two datasets. The blue dots in Figs. 2 and 3 represent the median popularity score in each year, and the upper and lower end of vertical lines represent the 75th and 25th percentile, respectively. In general, the popularity scores in both datasets experience an upward trend over time, with a relatively large variance. Most of the terms in the Gene editing dataset have a low popularity score before 2010, while the popularity score of the Transplant terms is higher in the early years.

4.2. Experimental setups

Experiments were conducted to compare the performance of different methods in predicting popularity scores of candidate topics since there is not a gold standard to compare emerging topics. On the other hand, the popularity score can be objectively measured and compared. It also reflects important characteristics of emerging topics.

4.2.1. Baselines

For performance comparison, we selected the methods that represent research topics with keywords as baselines. Other methods that use different topic representations, e.g., topic models and document clusters, were not considered because they can not be directed compared with the proposed approach. One machine learning method, two regression baselines, and two rule-based methods were employed in the experiment.

LightGBM: The machine learning baseline was LightGBM (Ke et al., 2017), which is a high-performance gradient boosting framework based on decision tree algorithms.

Linear Regression (LR) and Polynomial Regression (PR) models: Linear regression with an L2 regularization term was adopted as the first regression predictive baseline. The second one was the approach proposed by Asooja et. al. (2016), which relies on the polynomial regression model. Instead of using multivariate time series, this model predicts the target merely based on its historical values (i.e., univariate time series).

Rule-based models: A naive model that simply copies the value at the previous time step (t-1) as the prediction of current time step (t) was chosen as one baseline. The second rule-based method was Emergence Score (EScore) proposed by Porter et. al. (2019), which has been integrated into a text mining software, the Vantage Point, to help users identify emerging topics and related "players" from scientific or technological documents. The EScore is calculated based on the document frequency in different periods of the training data (Porter et al., 2019). The topics identified from the training data were ranked according to their EScore values. Thus, only the ranking performance was evaluated.

4.2.2. Training strategy: global vs. local

We trained neural network models to predict the popularity score in the next five years based on the nine features (including the popularity score itself) in the preceding eleven years. Specifically, the models were trained on the data between 2003 and 2013 to predict target indicators between 2014 and 2018. The research topics were first modeled with global strategy, in which one model was trained on all samples from a dataset. Next, with local strategy, we grouped the topics into different categories and trained separate models on each category, which allowed the training of fine-grained models. Models were tuned and evaluated with 5-fold cross-validation. 80% of the terms were used to train the models in each fold, while the other 20% were used for evaluation.

It should be noted that global and local training strategies were only applied to LSTM, NNAR, and LR. For PR, it was reported that

Table 2
Descriptive statistics of the two bibliographic datasets.

Dataset	Documents	Keywords	Venues	Authors	References	Identified topics
Gene editing	19,164	22,111	2,067	77,049	404,017	2,643
Transplant	206,573	59,026	25	437,501	831,163	5,141

Note: Identical entities (keyword, venue, author, and reference) were counted only once.

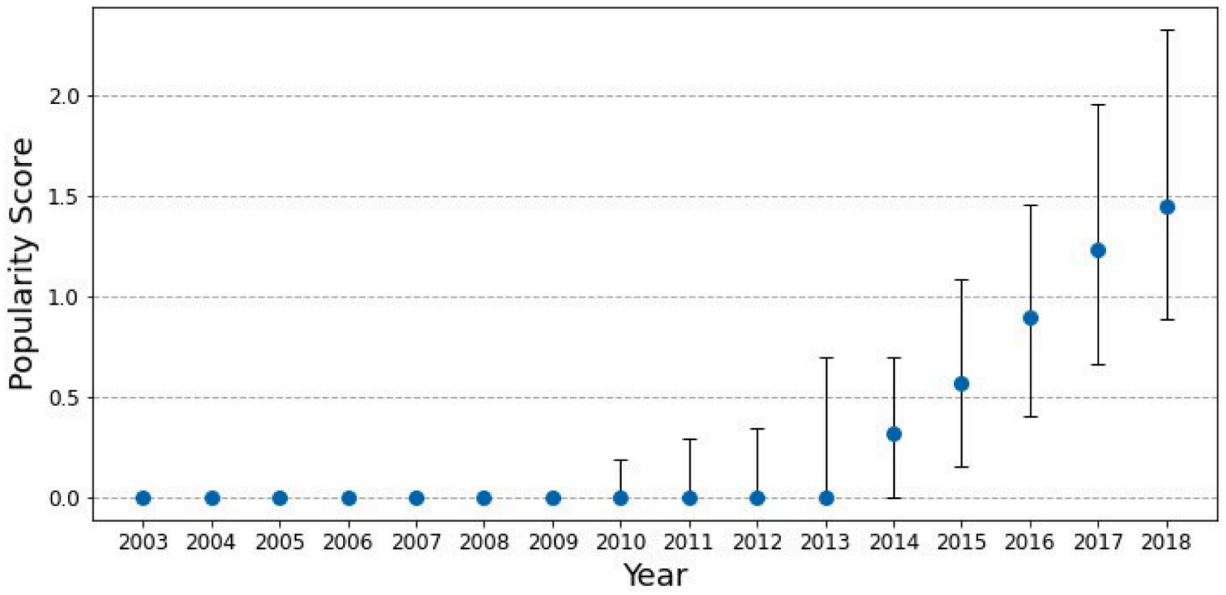


Fig. 2. Annual changes of the popularity scores in Gene editing dataset.

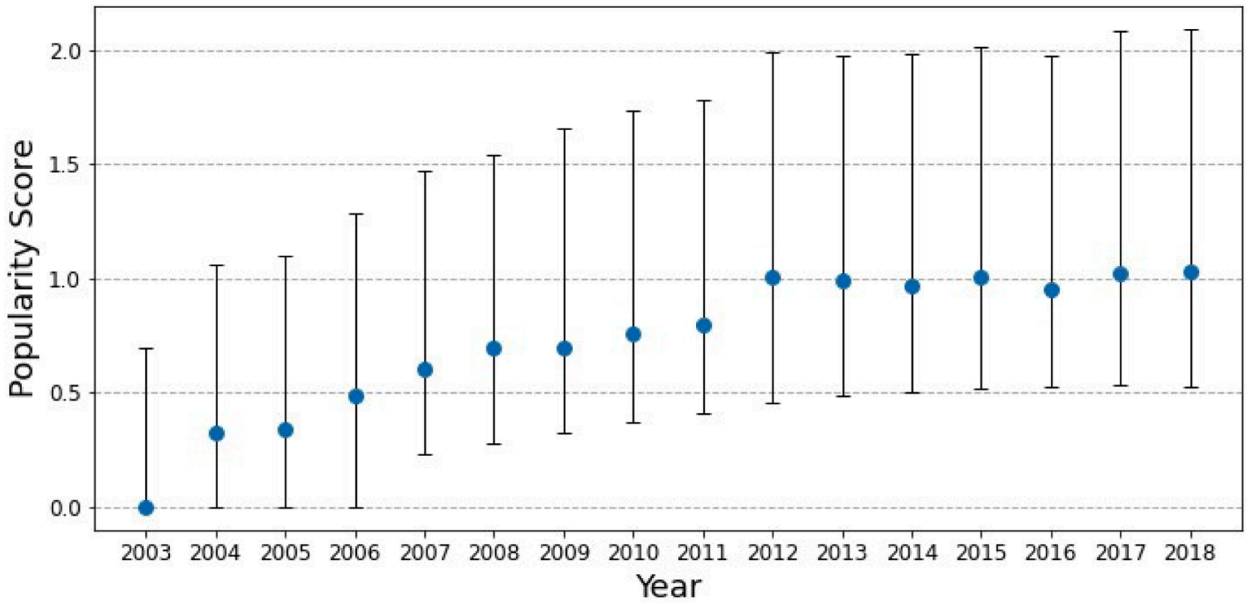


Fig. 3. Annual changes of the popularity scores in Transplant dataset.

individually modeling the time series data for every keyword achieved the best performance (Asooja et al., 2016). We followed this training process and compared the overall performance with other models. The rule-based models essentially predict the target indicator of individual topics, which do not apply any training strategy.

4.2.3. Evaluation metrics

We introduced two error-based and one rank-based metrics to evaluate the predictive models. While the error-based metrics measure the performance of predicting correct popularity scores, the rank-based metric reflects the model's capability to sort research topics according to their popularity scores correctly.

Mean absolute error (MAE). MAE measures the average distance between the predictions and ground truth. We calculated the annual and overall MAE metrics to measure the model performance as the following equations:

$$MAE_{t'} = \frac{1}{n}\sum_{i=1}^{n}\left|\hat{y}_{i,t'} - y_{i,t'}\right| \tag{3}$$

$$MAE_{overall} = \frac{1}{t'}\sum_{i=1}^{t'} MAE_{t'} \tag{4}$$

where n is the number of research topics, $\hat{y}_{i,t'}$ represents the predicted popularity score for topic i at time t', and $y_{i,t'}$ is the true popularity score of topic i at time t'. MAE is a straightforward metric for interpretation purposes as the errors are in the same unit as the prediction.

Root mean squared error (RMSE). While similar to MAE, RMSE penalizes large errors more heavily. The annual and overall RMSE are calculated as follows:

$$RMSE_{t'} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(\hat{y}_{i,t'} - y_{i,t'}\right)^2} \tag{5}$$

$$RMSE_{overall} = \frac{1}{t'}\sum_{i=1}^{t'} RMSE_{t'} \tag{6}$$

where n is the number of research topics, $\hat{y}_i$ represents the predicted popularity score for topic i, and y_i is the true popularity score of topic i.

Normalized discounted cumulative gain@k (NDCG@k). NDCG@k is a measurement of ranking quality according to the order of the top k results. It is often used in information retrieval to evaluate the performance of search engines and recommendation systems. The formulation of NDCG@k is given by:

$$DCG@k = \sum_{i=1}^{|pred@k|}\frac{2^{y_i} - 1}{log_2(i+1)} \tag{7}$$

$$IDCG@k = \sum_{i=1}^{|true@k|}\frac{2^{y_i} - 1}{log_2(i+1)} \tag{8}$$

$$NDCG@k = \frac{DCG@k}{IDCG@k} \tag{9}$$

where i represents the position number (starting from one) of the i-th topic and y_i denotes its true popularity score, *pred@k* represents the list of top k research topics ranked by their predicted popularity score, and *true@k* denotes the list of top k research topics ranked by their true popularity score. NDCG@k measures to what extent a model produces the ideal order of research topics. Higher NDCG@k represents better performance, ranging from zero to one. Specifically, we use NDCG@20 to measure the quality of the top 20 predictions in this study.

It should be noted that MAE and RSME are not applicable to the baseline of EScore method. Only NDCG@20 scores were obtained for EScore method.

4.3. Model tuning process

For both models, we fine-tuned the number of layers and the number of neurons/units per layer as they are the two most important hyperparameters of neural networks. The optimal hyperparameters of each model and dataset are presented in Table 3. The weights were updated using Adam optimizer, which computes individual adaptive learning rates for different parameters during training.

4.4. Evaluation results of predictive models with different strategies

4.4.1. Global strategy

Table 4 compares the five-year average performance of predictive models under the global strategy. LSTM and NNAR outperform

Table 3
Hyperparameters of neural network models.

Model	Gene editing		Transplant	
	Layers	Units	Layers	Units
NNAR	3	128	5	256
LSTM	5	512	1	512

Table 4
Overall evaluation results of the global strategy.

Model	Gene editing MAE	RMSE	NDCG@20	Transplant MAE	RMSE	NDCG@20
LSTM	**0.728**	**1.279**	0.563	**0.754**	**1.244**	0.509
NNAR	0.729	1.284	0.454	0.766	1.266	0.478
LightGBM	0.821	1.334	**0.610**	0.821	1.302	**0.563**
LR	0.802	1.360	0.249	0.830	1.308	0.389
PR (Asooja et. al., 2016)	1.182	1.988	0.005	1.787	2.662	0.051
Naïve	0.995	1.662	0.437	0.981	1.618	0.081
EScore (Porter et. al., 2019)	-	-	0.484	-	-	0.322

other models in terms of MAE and RMSE. LSTM also achieves a high score of NDCG@20. Although LightGBM has inferior MAE and RMSE performance, it obtains the highest NDCG@20 score in both datasets. It is interesting that this tree-based machine learning model is able to predict more accurate results for top terms, even though it produces more errors than neural networks in general.

The approach proposed by Asooja et. al. (2016) performs poorly in our experiment, especially for the ranking metric. Without enough samples and multivariate time series, this approach is likely to overfit individual patterns in the training period and underperform in the test period. With global strategy, LR shows better performance than PR. The result reflects that there may be collective patterns of term usage in a research domain. Training a global model may improve the overall performance of predicting popularity scores for topics.

Surprisingly, the rule-based method, EScore, performs relatively well in terms of NDCG@20. It is even better than the NNAR model on the Gene editing dataset. The naïve model also shows good performance in the Gene editing dataset in terms of NDCG@20. It shows that the topics with high impact and growth rate may last for years, which may help the identification of future emerging topics. It also suggests that human-defined rules may be useful in the pipeline of emerging topic prediction.

Overall, the LSTM sequence-to-sequence model achieves the best results, indicating its ability to capture the non-linear pattern of topic development. Due to the localized nature and inferior performance of the approaches proposed by Porter et al. (2019) and Asooja

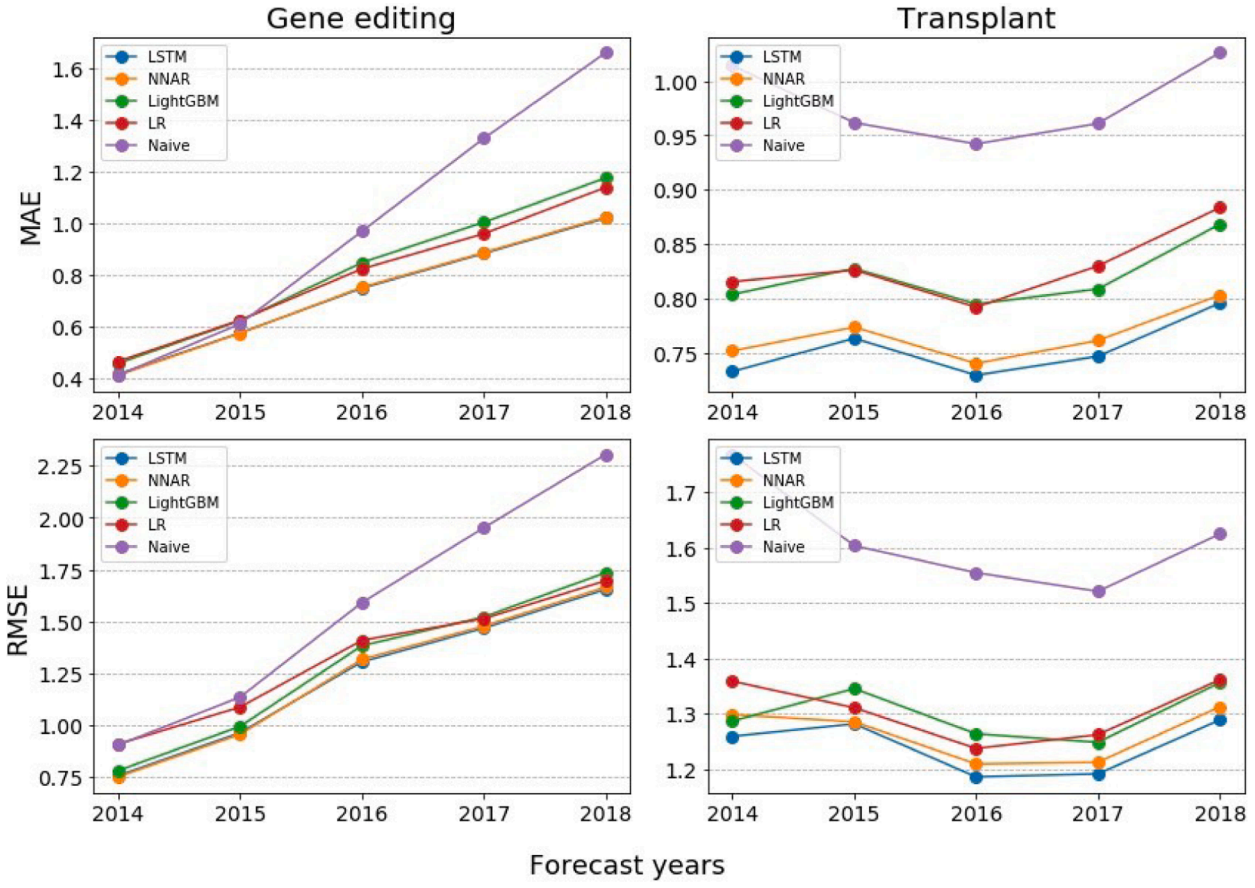


Fig. 4. Annual evaluation results of the global strategy.

et al. (2016), we do not include them in further analyses.

Fig. 4 shows the annual model performance in different forecast years. MAE and RMSE increase significantly over the five-year period on the Gene editing dataset, indicating that the prediction for farther time is more difficult than the recent ones. However, the performance is more stable on the Transplant dataset, and the difference between the five models is more obvious on it. The performances of two neural network models are very close on both datasets. The error of the naive model increases sharply after the second year (2015) on Gene editing.

4.4.2. Local strategies

Different research topics may have various patterns in their popularity scores over time. We considered two types of local training strategies that separate research topics into different groups. In the first local strategy, terms were divided into three groups (high, middle, and low) according to the percentile of total popularity scores which were summed up over the eleven years of the training dataset. One separate model was trained on each term group, then was used to predict the future popularity score of the terms in the group. Alternatively, the second local strategy grouped terms according to the changing trend (upward vs. downward) of popularity score. The trend of a term was measured by the coefficient of the linear regression model fitted to its popularity scores in the eleven years.

Table 5 shows the performance of different models with the two local strategies. Regarding the first local strategy, LSTM outperforms other models in most of the metrics. LSTM also obtains a higher NDCG@20 score than LightGBM on the Gene editing dataset. Compared with the global strategy (Table 4), LSTM has higher NDCG@20 scores on the Gene editing dataset while the other metrics are slightly worse. On the Transplant dataset, LSTM outperforms other models in terms of all metrics except NDCG@20.

As for the second local strategy, most of the models are worse than those with the global strategy. LSTM still ranks top in all metrics except RMSE on the Gene editing dataset. Surprisingly, the NDCG@20 performance scores of LightGBM with both local strategies are much worse than with the global strategy, while its MAE and RMSE performances increase. The changes in performances of NNAR are marginal. Therefore, we conclude that both local strategies do not improve the overall model performance over the global strategy in this study.

To sum up, neural networks outperform other baseline models in terms of most metrics on both datasets with two types of strategies. And LSTM performs better than NNAR. It is interesting that LightGBM can obtain a high NDCG@20 score, while its MAE and RMSE are relatively large with the global strategy.

4.4.3. Analysis of prediction results

We further compared the predicted popularity scores by LSTM and LightGBM with the scores of ground truth, since they were the best models in terms of error-based and rank-based evaluation metrics, respectively. Fig. 5 shows the results of eight research topics by LSTM and LightGBM with global strategy. The four samples in the first row represent the high-score topics whose total popularity scores are above average in the forecast period, while the ones in the second row are with low scores. In terms of the high-score topics, they generally demonstrate the potentials of becoming emerging topics in early years. Both LSTM and LightGBM are capable of predicting the changes of popularity score, while the predictions given by LSTM are closer to the ground truth. However, they behave differently when predicting the low-score topics. LSTM becomes more conservative and makes smooth predictions that reflect the general trend. By contrast, the prediction results of LightGBM tend to be more fluctuated and less accurate. This is possibly caused by the different mechanisms of producing multi-step predictions used by the two models. LSTM predicts recursively, affected by previous predictions, while LightGBM produces multiple independent outputs at once (Ben Taieb, Bontempi, Atiya, & Sorjamaa, 2012).

4.5. Emerging topics nominated by different methods

We trained the NN models and LightGBM on the whole dataset (from 2008 to 2018) with the global strategy, and used them to predict popularity scores between 2019 and 2023 for the candidate topics. The results contain the topics with high predicted popularity scores, which reflect that the topics have a large impact and grow fast. Apart from the two properties, emerging topics should also

Table 5
Overall evaluation results of the local strategies.

Model	Gene editing MAE	RMSE	NDCG@20	Transplant MAE	RMSE	NDCG@20
Local strategy #1						
LSTM	**0.729**	**1.284**	**0.603**	**0.756**	**1.249**	0.490
NNAR	0.734	1.286	0.458	0.767	1.268	0.485
LightGBM	0.796	1.288	0.456	0.824	1.299	**0.543**
LR	0.804	1.378	0.277	0.828	1.323	0.373
Naïve	0.995	1.662	0.437	0.981	1.618	0.081
Local strategy #2						
LSTM	**0.728**	1.286	**0.551**	**0.753**	**1.247**	**0.508**
NNAR	0.729	1.287	0.467	0.766	1.271	0.485
LightGBM	0.791	**1.278**	0.433	0.816	1.283	0.489
LR	0.805	1.371	0.237	0.842	1.342	0.394
Naïve	0.995	1.662	0.437	0.981	1.618	0.081

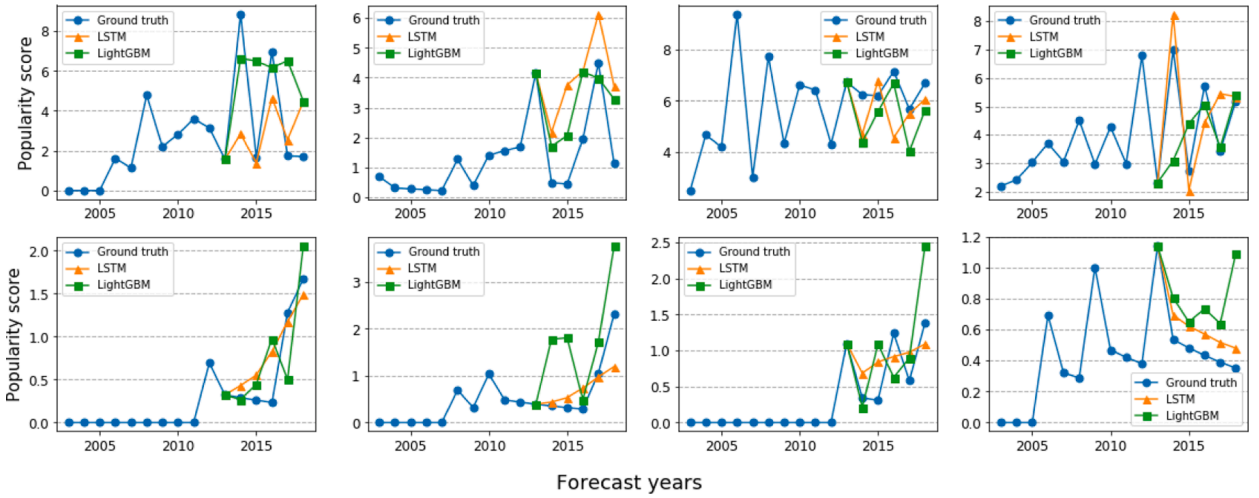


Fig. 5. Comparison of the predictions given by LSTM and LightGBM.

be novel. We then filtered the topics with the novelty indicator. In total, we filtered out 118 (out of 2,643) and 870 (out of 5,141) novel topics for the Gene editing and Transplant dataset, respectively. Emerging topics can be ranked and selected according to the predicted popularity scores.

Table 6 presents the top 20 emerging topics of Gene editing nominated by LSTM, NNAR, LightGBM, and Escore. In general, the emerging topics nominated by machine learning models are more alike compared with those given by the rule-based model. LSTM and NNAR share 14 research topics, while LSTM, NNAR, and LightGBM have only 8, 8 and 4 common topics with EScore, respectively. Four

Table 6
Top 20 emerging research topics of Gene editing nominated by four models.

Rank	LSTM	NNAR	LightGBM	EScore
1	Clustered regularly interspaced short palindromic repeats (CRISPR) *	Clustered regularly interspaced short palindromic repeats (CRISPR) *	Zinc-finger nucleases (ZFN) *	Targeted gene alteration *
2	Transcription activator-like effectors nucleases (TALEN) *	Transcription activator-like effectors nucleases (TALEN) *	Keratopathy +	Inverted repeat *
3	T cell #	Genetic modification *	Clustered regularly interspaced short palindromic repeats (CRISPR) *	Mycobacterium tuberculosis complex (MTC) #
4	Definitive endoderm	T cell #	Motor neurons #	Clustered regularly interspaced short palindromic repeats (CRISPR) *
5	guide RNA (gRNA) #	Zinc-finger nucleases (ZFN) *	Transcription activator-like effectors nucleases (TALEN) *	Integrase-defective lentiviral vectors (IDLVs) *
6	Genetic modification *	guide RNA (gRNA) #	guide RNA (gRNA) #	CpG oligodeoxynucleotides (ODN) #
7	Zinc-finger nucleases (ZFN) *	Definitive endoderm	Neurodegenerative disease +	Definitive endoderm
8	Nonhomologous end joining (NHEJ) *	Nonhomologous end joining (NHEJ) *	Genetic modification *	Herpes simplex virus (HSV)
9	Neurodegenerative disease +	Neurodegenerative disease +	T cell #	T cell #
10	Homology-directed recombination (HDR) *	Motor neurons #	Green fluorescent protein #	Motor neurons #
11	Acetolactate synthase	Keratopathy +	Xanthomonas #	Glanzmann thrombasthenia +
12	Pulmonary hypertension +	Homology-directed recombination (HDR) *	Spermatogenesis	Infraorbital nerve
13	Cancer cell lines #	Transfection efficiency	Nonhomologous end joining (NHEJ) *	Reverse transcriptase-polymerase chain reaction (RT-PCR) *
14	Gastric cancer +	Spermatogenesis	Heat-stable nucleoid structuring	Suppressor phenotype *
15	Mesenchymal stem cell (MSC) #	Enhanced green florescent protein #	Pulmonary hypertension +	Nonhomologous end joining (NHEJ) *
16	Suppressor phenotype *	Glanzmann thrombasthenia +	Lactic acid bacteria #	Organellar genome
17	Transfection efficiency	Duchenne muscular dystrophy +	Acetolactate synthase	Sequence type
18	Motor neurons #	Sequence type	Condyloma acuminatum #	MRE11 (gene) #
19	Keratopathy +	Myostatin gene #	Enhanced green fluorescent protein #	Cancer cell lines #
20	Reverse transcriptase-polymerase chain reaction (RT-PCR) *	Cancer cell lines #	Homology-directed recombination (HDR) *	Peptide nucleic acid (PNA) #

Note: * denotes technologies, + denotes diseases, and # denotes cell lines

topics appear in all the topic lists, including clustered regularly interspaced short palindromic repeats (CRISPR), nonhomologous end joining (NHEJ), motor neurons, and T cell.

For the validation purpose, we manually labeled the nominated topics in Table 6 with the corresponding classification, which includes technology, disease, and cell lines. A literature review is also conducted to survey the importance and possible developments of these topics. Among them, the most important topics are ZFN, TALEN, and CRISPR, representing three revolutionary technologies that enable effective genome editing (Hill et al., 2014). All three technologies appear in the top 20 topics nominated by machine learning models but in a different order. Generally, genome editing performs knock in and out operations on genetic sites and makes use of the HDR and NHEJ pathways to change DNA sequences. Different from nuclease-based technologies (ZFN and TALEN), CRISPR is based on guide RNA, which is simpler and more efficient. Therefore, CRISPR has gained increasing attention from scientists and its inventors were awarded the 2020 Nobel Prize in Chemistry (The Royal Swedish Academy of Sciences, 2020). Recently, genetic approaches have also been proposed to early detect and cure intractable diseases like cancers, keratopathy, and condyloma acuminatum. They are also nominated by our methodology. More intriguingly, a technology nominated by the LSTM model, RT-PCR, has become the most important way to test COVID-19 nowadays (Fang et al., 2020).

Table 7 shows the top 20 research topics of Transplant nominated by the three models. LSTM and NNAR share 11 topics, LSTM, NNAR, and LightGBM share 5, 6 and 6 topics with EScore, respectively. In addition, four topics are nominated by all models, including acute kidney injury, T cell-mediated rejections, extracorporeal membrane oxygenation (ECMO), and ex vivo lung perfusion (EVLP).

With respect to the research topics in the Transplant dataset, kidney-related topics are among the most important ones, as kidney is the most commonly transplanted organ. Acute kidney injury (AKI) is a fatal disease that requires kidney transplantation but it is also a common disease that happens on transplant recipients. Although a number of scientists have devoted themselves to study AKI, its broad causes and deadly complications make AKI still a difficult problem that deserves more attention. Recently, scientists have found that it is also a severe complication of COVID-19 (Batlle et al., 2020). It is nominated by all four approaches and has a high rank. In addition, technologies such as EVLP and stem cell transplantation are also nominated as they are promising cures for the diseases that

Table 7
Top 20 emerging research topics of Transplant nominated by four models.

Rank	LSTM	NNAR	LightGBM	EScore
1	Acute kidney injury +	Acute kidney injury +	Neutrophil gelatinase-associated lipocalin #	T cell-mediated rejections +
2	Device thrombosis +	Gastroesophageal varices	Self-management	Extracorporeal membrane oxygenation (ECMO) *
3	Extracorporeal membrane oxygenation (ECMO) *	Extracorporeal membrane oxygenation (ECMO) *	Donation after brain death (DBD) *	Acute kidney injury +
4	Adipose tissue-derived stem cells #	T cell-mediated rejections +	Clofarabine △	Self-management
5	Galactosyl transferase gene knockout *	Device thrombosis +	T cell-mediated rejections +	Extended release tacrolimus capsules △
6	Ex vivo lung perfusion (EVLP) *	Ex vivo lung perfusion (EVLP) *	Hematopoietic cell transplantation-specific comorbidity index (HCT-CI)	Prolonged-release tacrolimus △
7	Donation after brain death (DBD) *	De novo donor-specific antibody #	Acute kidney injury +	Metrics
8	T cell-mediated rejections +	Hematopoietic cell transplantation-specific comorbidity index (HCT-CI)	Adipose tissue-derived stem cells #	Right heart catheterization (RHC) *
9	Mean Fluorescence Intensity (MFI)	Seattle heart failure model *	Extracorporeal membrane oxygenation (ECMO) *	Ex vivo lung perfusion (EVLP) *
10	Mammalian target of rapamycin inhibitors (mTORi) △	Neutrophil gelatinase-associated lipocalin #	Galactosyl transferase gene knockout *	Temporary circulatory support *
11	Hematopoietic cell transplantation-specific comorbidity index (HCT-CI)	Calculated panel-reactive antibody (CPRA)	Ex vivo lung perfusion (EVLP) *	Gastroesophageal varices
12	Organ care system *	Human leukocyte antigen DQ #	Kidney transplantation patient	Within-patient variability
13	Kidney transplantation patient	Veno venous extracorporeal membrane oxygenation (VV-ECMO) *	Mean Fluorescence Intensity (MFI)	Chronic thromboembolic pulmonary hypertension (CTEPH) *
14	Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI)	Galactosyl transferase gene knockout *	De novo donor-specific antibody #	Tricuspid annulus plane systolic excursion (TAPSE)
15	Neutrophil gelatinase-associated lipocalin #	Hepatic portal vein	Autologous hematopoietic cell transplant *	MELD-Na score
16	Self-management	Donation after brain death (DBD) *	Calculated panel-reactive antibody (CPRA)	Solid-organ transplant recipients
17	Autologous hematopoietic cell transplant *	Clofarabine △	Incidence rate ratio	Nanofibrous scaffold *
18	De novo donor-specific antibody #	Hepatic venous outflow obstruction +	Device thrombosis +	Decitabine △
19	B-cell-activating factor (BAFF)	Sotrastaurin △	Mammalian target of rapamycin inhibitors (mTORi) △	Percutaneous transhepatic biliary drainage (PTBD) *
20	Calculated panel-reactive antibody (CPRA)	Chronic thromboembolic pulmonary hypertension (CTEPH) *	Chronic thromboembolic pulmonary hypertension (CTEPH) *	Transplant associated thrombotic microangiopathy

Note: * denotes technologies, + denotes diseases, # denotes cell lines, and △ denotes drugs

used to be terminal illnesses, e.g., lung cancer and leukemia (Machuca et al., 2015). Compared with the Gene editing dataset, emerging research topics of Transplant also includes drugs and indices. For example, mTOR is an important kinase that regulates cell growth, proliferation, metabolism, and survival. Its increased activation is observed in patients with transplant rejection and its inhibitors (i.e., mTORi) have been introduced to prevent renal allograft rejection. More recently, mTOR has been found on a critical pathway to the growth and reproduction of cancer cell lines. Therefore, mTOR inhibitors such as Everolimus and Temsirolimus have been repurposed to the treatment of various tumors (Hua et al., 2019). It is worth noting that ECMO, a life-support machine for transplantation patients, is now serving as an important technology to support COVID-19 patients (Savarimuthu, BinSaeid, & Harky, 2020).

Through the above qualitative analysis, we conclude that the proposed methodology is able to nominate emerging research topics.

5. Discussion

This study proposed a two-step solution to tackle the task of emerging research topic prediction. Our solution takes the growth, impact, and novelty of research topics into consideration and employs a hybrid approach of machine learning and bibliometric indicator. The prediction of popularity score that measures both growth and impact is formulated as a multivariate multi-step time series forecasting problem and solved by machine learning models, while the novelty of emerging topics is ensured by the bibliometric indicator. Experiments on two datasets show the effectiveness of our approach. In this section, we first discuss the influence of the decay factor on the performance of our solution, then summarize the key findings and their implications. Limitations of our study and future work are presented as well.

5.1. Effects of the decay factor

As a predefined parameter in the formula of popularity score (Section 3.2.2), the decay factor α controls the extent to which the historical impact of a research topic influences the impact at present. A larger decay factor leads to an overall increase in the popularity

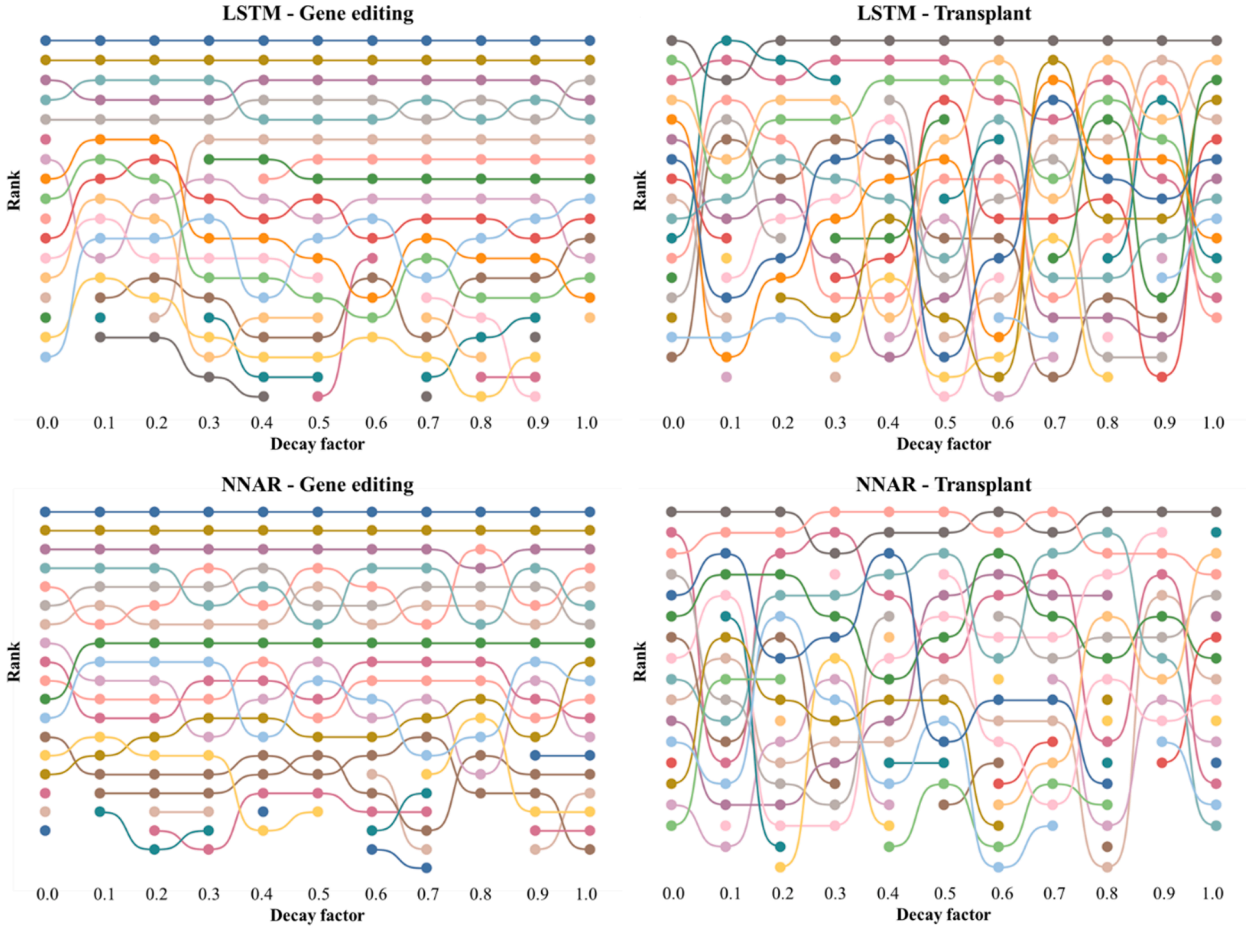


Fig. 6. The top 20 predicted emerging topics with the highest popularity scores using different decay factor values. Each column of the bump chart represents a ranked list of emerging topics predicted by the neural network models with a specific value of decay factor. The links between nodes of the same color represent the rank change of a research topic over different decay factors.

score. To observe how the factor influences the predicted topics, we compared the top 20 research topics with the highest popularity scores generated by LSTM and NNAR using different decay factor values. The factor varied from 0.0 to 1.0, with a step of 0.1. Fig. 6 shows the results, in which nodes of the same color denote the same research topic nominated by the models with different decay factors. There are also links connecting the same topics over different decay factors to keep track of their changes in ranks.

For Gene editing, a total of 30 research topics are nominated by 11 LSTM models trained with different values of decay factor. Among them, 10 topics appear in the results of all trained models. It is also shown in Fig. 6 that the rank of the emerging topics over different decay factors is relatively stable, especially for the top 5 topics. For Transplant, only 30 topics are nominated by 11 LSTM models. Given that there are 870 novel Transplant research topics, the nominated topics indeed account for a small and stable fraction. Although the topic rank over different decay factors is not as stable as that of the Gene editing dataset, 9 research topics are shared in the results of all models. In addition, the rank of one topic, acute kidney injury, is consistently the first in all models except the one with a decay factor of 0.1. Regarding the NNAR method, all 11 models with different decay factors only nominated 29 emerging topics, among which 14 topics are shared by all models on Gene editing. The 11 NNAR models nominated 41 emerging topics, among which 6 topics are common ones. The above results indicate that our methodology is robust across different values of decay factors.

5.2. Key findings

A few findings are obtained through the experiments.

First, it is shown that neural networks outperform other baseline models in terms of most metrics on both datasets with two types of strategies. LSTM performs better than NNAR, while LightGBM outperforms LSTM in terms of the rank-based metric (NDCG@20) with the global training strategy. In other words, LightGBM can rank the topics of high popularity score better and the LSTM sequence-to-sequence model can predict more accurate popularity scores for all topics in general. All machine learning models outperform the rule-based naïve model significantly, indicating that this problem can be effectively modeled as a machine learning task. More sophisticated models could be proposed by future studies to improve performance.

Second, there is no significant difference between the model performances using different training strategies. All models achieved better results in terms of MAE and NDCG@20 on the Gene editing dataset, which was smaller in sample size and less challenging, than on the Transplant dataset. If we further separated each dataset according to popularity score and developing trend of research topics, the performances of most models slightly declined except LightGBM, which achieved better MAE and RMSE but lower NDCG@20.

In addition, further analysis of the model predictions for individual topics shows that LightGBM tended to make more fluctuated predictions, which may be caused by its nature of independent multiple outputs. As a comparison, LSTM made more accurate and smooth predictions for low-score topics.

Last, we found that the emerging topics nominated by our two-step solution include some important topics in the field, including recently awarded Nobel prizes. Without a gold standard on what the emerging topics of the fields should be, our literature analysis of the nominated topics shows promise.

5.3. Theoretical and practical implications

Overall, the implications of this study are twofold.

Theoretically, this study sheds valuable light on emerging research topic prediction by combining machine learning and bibliometric indicator. We initially attempted to model the whole problem of emerging topic prediction as a time series forecast problem, in which an appropriate target indicator should take the properties of impact, growth and novelty into consideration. However, we only manage to capture impact and growth by proposing the indicator of popularity score. It is difficult to embed all three significant properties into a sole indicator. In this study, predicting future emerging research topics is transformed into two subtasks, i.e., predicting future popularity scores of topics and filtering novel ones. The machine learning framework is mainly applied to solve the first subtask, which could be formally formulated as a statistical learning problem.

The lessons learned from the machine learning approach to popularity score prediction can enlighten future solutions to the subtask. The machine-learning paradigm, including candidate topics generation, feature extraction, model training and prediction, has been evidenced as a feasible approach. Different machine learning methods can be compared on this subtask.

Practically, the present work provides a viable approach to predict the future emerging research topics for various stakeholders, including but not limited to researchers, policymakers, and librarians. Our proposed two-step solution can be used to guide the construction of emerging topic prediction systems. The term-based topic representation and metadata-based features enable the system adaptable to new data, which will help monitor the change of emerging topics in real time. Our study has shown the effectiveness of LSTM and NNAR. Other neural network models can further be applied to optimize the system under the proposed framework. In addition, the prediction of target indicators can help researchers understand the evolutionary trajectories of topics.

For predictive models, it is shown that neural networks outperform other baseline models. A possible explanation for this is that the relationships between the features of topics and their emergence measured by indicators are not linear, which can be well captured by non-linear neural networks.

When training the models, the global strategy could be favored since local strategies that train multiple models with different groups of terms do not improve the performance, which is different from the result in Asooja et al. (2016). One possible reason is that machine learning methods like neural networks need a large number of training samples, while linear and polynomial regressions require less training data. Typically, increasing the sample size will help neural networks achieve better results and prevent overfitting. Therefore, the global strategy that leverages more samples generally benefits model performance.

5.4. Limitations and future work

There are some limitations in this study. The datasets we employed for model training and prediction were domain-specific; therefore, it remains unclear whether emerging topics in a multidisciplinary scenario share a similar pattern to the ones in a specific research field. More importantly, two issues related to problem scope and solution approach need further discussions and investigations to deepen the understanding of emerging research topic prediction.

Firstly, the scope of most extant studies, including this research, is limited to the existing research topics. Although it is practical to predict emerging topics from existing ones because they have already left traces in the historical data, there is a possibility that brand new topics will emerge and become popular in the future, especially when the prediction period is long. Novel topics may also be derived from the combination of existing research topics. Instead of the indicator of a single topic, modeling the interaction between research topics is also important. It is possible for future studies to formulate this problem as link prediction, which directly models and forecasts the interactions among topics.

Secondly, this study treats emerging research topic prediction as a regression problem, which is also a common practice in existing studies (McKeown et al., 2016; Xu et al., 2019). However, the classification approach, which classifies research topics as emerging or not based on historical features, is seldom adopted by researchers. Since there are no widely accepted ground truth or gold standard for determining emerging topics (Small et al., 2014; Wang, 2018), the large labeled dataset required to train classifiers is unavailable. Classification may be another promising approach to solve this problem if widely recognized criteria of emerging research topics are proposed. For instance, the research of winners of academic awards like the Nobel Prize may be utilized as the proxy of emerging research (Small et al., 2014).

6. Conclusion

Identifying and predicting emerging research topics are important for various stakeholders, for both research and management purposes. While many scholars have attempted to identify emerging topics retrospectively, few efforts have been made to the prediction work with machine learning approaches. This study provides a two-step solution that combines machine learning and bibliometric indicator approaches. The first step models emerging research topics prediction as a multivariate multi-step time series forecasting problem and predicts the popularity scores of candidates. The second step selects novel topics from the candidates from the first step. This study makes theoretical contributions to emerging research topic prediction and can guide the practice of related stakeholders. Future work may focus on the prediction for new topics and the application of classification models to this problem.

CRediT authorship contribution statement

Zhentao Liang: Methodology, Software, Writing – original draft, Formal analysis. **Jin Mao:** Methodology, Investigation, Writing – review & editing. **Kun Lu:** Investigation, Writing – review & editing. **Zhichao Ba:** Investigation. **Gang Li:** Supervision, Funding acquisition.

Declaration of Competing Interest

No.

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